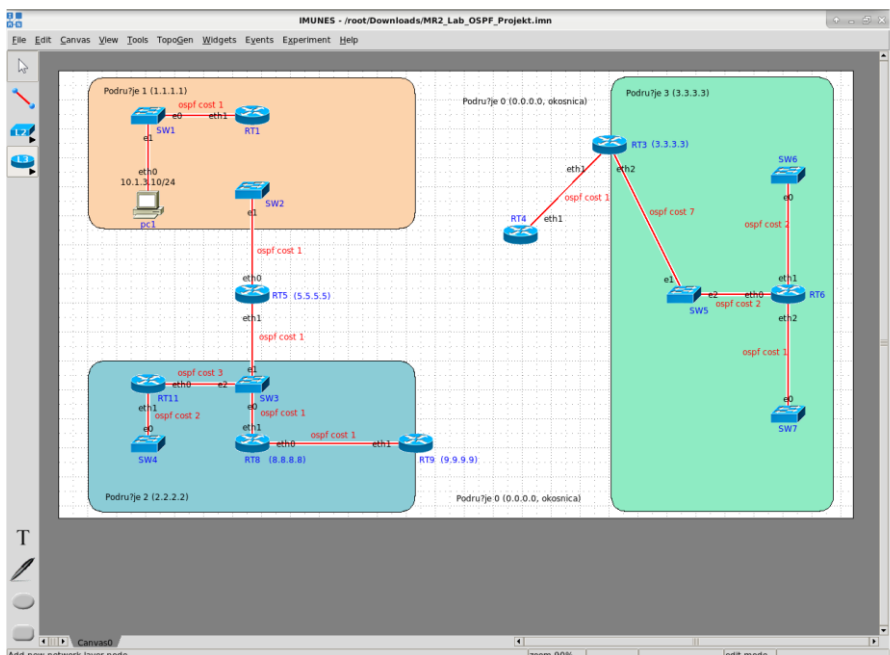


Lab 6: Usmjeravanje u većim mrežama – hijerarhijski OSPF

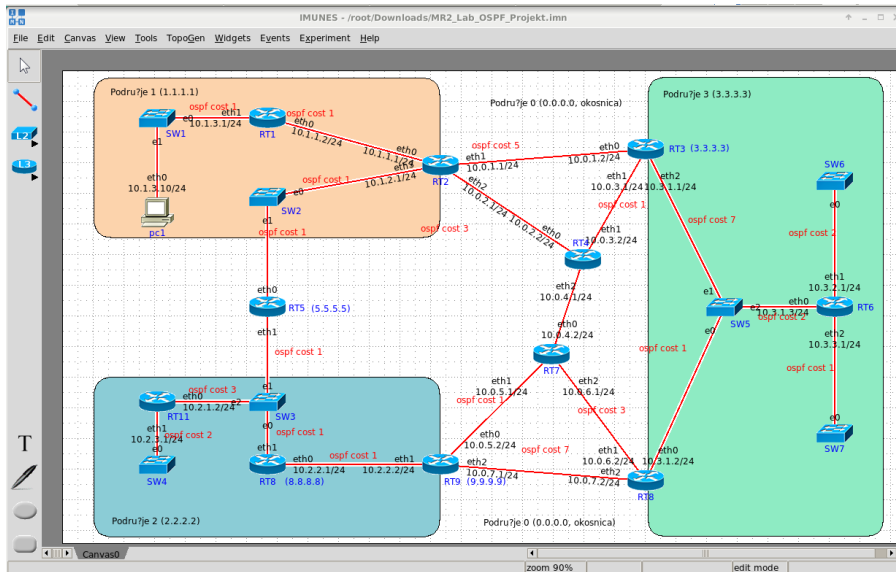
Kreiranje novog korisnika david naredbom `adduser` :

```
xterm
[root@IMUNES ~]# adduser david
Username: david
Full name: David Slavik
Uid (Leave empty for default):
Login group [david]:
Login group is david, Invite david into other groups? []:
Login class [default]:
Shell (sh csh tcsh git-shell bash rbash nologin) [sh]:
Home directory [/home/david]:
Home directory permissions (Leave empty for default):
Use password-based authentication? [yes]:
Use an empty password? (yes/no) [no]: no
Use a random password? (yes/no) [no]: no
Enter password:
Enter password again:
Lock out the account after creation? [no]: no
Username      : david
Password      : *****
Full Name     : David Slavik
Uid           : 1002
Class         :
Groups        : david
Home          : /home/david
Home Mode     :
Shell         : /bin/sh
Locked        : no
OK? (yes/no): y
OK? (yes/no): y
adduser: INFO: Successfully added (david) to the user database.
Add another user? (yes/no): n
Goodbye!
[root@IMUNES ~]#
```

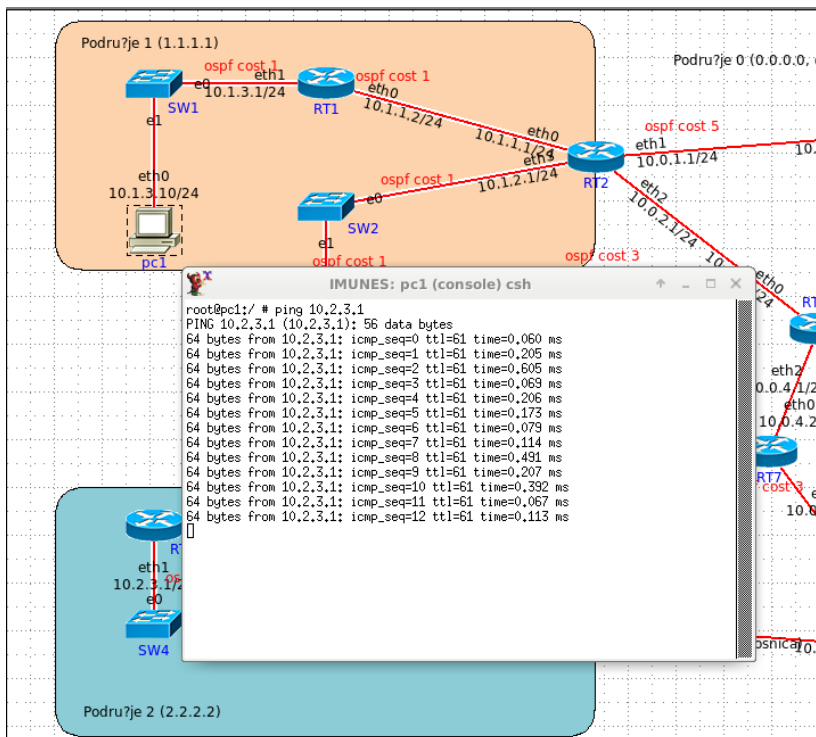
Otvaranje projekta koji je potrebno dovršiti u IMUNES-u:



Dopunjena verzija projekta:



Naredba **ping** prema 10.2.3.1:



Naredba **ping** prema 10.3.2.1:

```
IMUNES: pc1 (console) csh
64 bytes from 10.2.3.1: icmp_seq=56 ttl=61 time=0,156 ms
64 bytes from 10.2.3.1: icmp_seq=57 ttl=61 time=0,192 ms
64 bytes from 10.2.3.1: icmp_seq=58 ttl=61 time=0,246 ms
64 bytes from 10.2.3.1: icmp_seq=59 ttl=61 time=0,199 ms
64 bytes from 10.2.3.1: icmp_seq=60 ttl=61 time=0,259 ms
64 bytes from 10.2.3.1: icmp_seq=61 ttl=61 time=0,070 ms
64 bytes from 10.2.3.1: icmp_seq=62 ttl=61 time=0,116 ms
64 bytes from 10.2.3.1: icmp_seq=63 ttl=61 time=0,450 ms
64 bytes from 10.2.3.1: icmp_seq=64 ttl=61 time=0,252 ms
^C
--- 10.2.3.1 ping statistics ---
65 packets transmitted, 65 packets received, 0,0% packet loss
round-trip min/avg/max/stddev = 0,060/0,233/0,708/0,150 ms
root@pc1:/ # ping 10.3.2.1
PING 10.3.2.1 (10.3.2.1): 56 data bytes
64 bytes from 10.3.2.1: icmp_seq=0 ttl=60 time=0,065 ms
64 bytes from 10.3.2.1: icmp_seq=1 ttl=60 time=0,070 ms
64 bytes from 10.3.2.1: icmp_seq=2 ttl=60 time=0,077 ms
64 bytes from 10.3.2.1: icmp_seq=3 ttl=60 time=0,123 ms
64 bytes from 10.3.2.1: icmp_seq=4 ttl=60 time=0,338 ms
64 bytes from 10.3.2.1: icmp_seq=5 ttl=60 time=0,411 ms
64 bytes from 10.3.2.1: icmp_seq=6 ttl=60 time=0,095 ms
64 bytes from 10.3.2.1: icmp_seq=7 ttl=60 time=0,129 ms
```

Naredba **tracert** na eth0 sučelje:

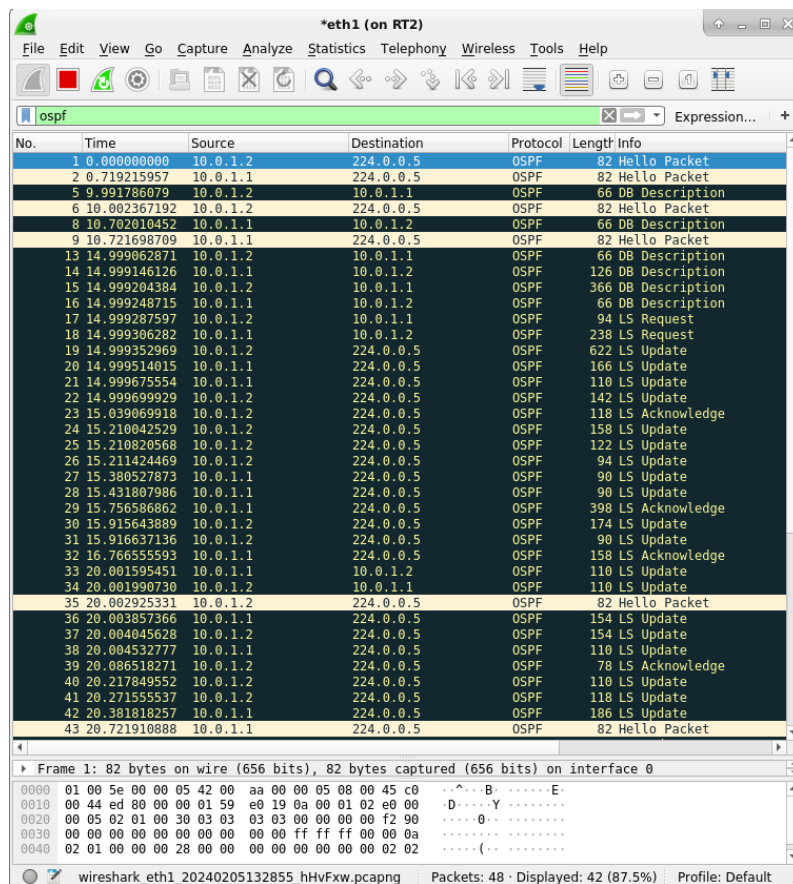
```
IMUNES: pc1 (console) csh
root@pc1:/ # tracert 10.2.1.2
tracert to 10.2.1.2 (10.2.1.2), 64 hops max, 40 byte packets
 1  10.1.3.1 (10.1.3.1)  0,193 ms  0,111 ms  0,100 ms
 2  10.1.1.1 (10.1.1.1)  0,131 ms  0,112 ms  0,103 ms
 3  10.1.2.2 (10.1.2.2)  0,141 ms  0,115 ms  0,109 ms
 4  10.2.1.2 (10.2.1.2)  0,143 ms  0,122 ms  0,112 ms
root@pc1:/ #
```

Rezultat je očekivan jer je to najkraći put (na temelju OSPF cijena).

Za daljnji nastavak vježbe, postavljena je OSPF cijena između **RT5** i **SW3** na visoku vrijednost, **100**. Za očekivati je da će se uzeti drugačiji put jer je cijena puta znatno povećana (sigurno postoji jeftiniji put). To se i ostvarilo (slika u nastavku): vidljiv je novi put i 7 skokova umjesto prethodnih 4 jer novi nađeni put ima manju cijenu.

```
IMUNES: pc1 (console) csh
root@pc1:/ # tracert 10.2.1.2
tracert to 10.2.1.2 (10.2.1.2), 64 hops max, 40 byte packets
 1  10.1.3.1 (10.1.3.1)  0,179 ms  0,096 ms  0,102 ms
 2  10.1.1.1 (10.1.1.1)  0,133 ms  0,115 ms  0,104 ms
 3  10.0.3.2 (10.0.3.2)  0,212 ms  0,118 ms  0,163 ms
 4  10.0.4.2 (10.0.4.2)  0,142 ms  0,123 ms  0,113 ms
 5  10.0.5.2 (10.0.5.2)  0,080 ms  0,125 ms  0,161 ms
 6  10.2.1.1 (10.2.1.1)  0,152 ms  0,127 ms  0,116 ms
 7  10.2.1.2 (10.2.1.2)  0,187 ms  0,127 ms  0,152 ms
root@pc1:/ #
```

Pokretanje Wiresharka na usmjerniku RT2:



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.1.2	224.0.0.5	OSPF	82	Hello Packet
2	0.719215957	10.0.1.1	224.0.0.5	OSPF	82	Hello Packet
5	9.991786879	10.0.1.2	10.0.1.1	OSPF	66	DB Description
6	10.002367192	10.0.1.2	224.0.0.5	OSPF	82	Hello Packet
8	10.702010452	10.0.1.1	10.0.1.2	OSPF	66	DB Description
9	10.721698709	10.0.1.1	224.0.0.5	OSPF	82	Hello Packet
13	14.999062871	10.0.1.2	10.0.1.1	OSPF	66	DB Description
14	14.999146126	10.0.1.1	10.0.1.2	OSPF	126	DB Description
15	14.999204384	10.0.1.2	10.0.1.1	OSPF	366	DB Description
16	14.999248715	10.0.1.1	10.0.1.2	OSPF	66	DB Description
17	14.999287597	10.0.1.2	10.0.1.1	OSPF	94	LS Request
18	14.999306282	10.0.1.1	10.0.1.2	OSPF	238	LS Request
19	14.999352969	10.0.1.2	224.0.0.5	OSPF	622	LS Update
20	14.999514015	10.0.1.1	224.0.0.5	OSPF	166	LS Update
21	14.999675554	10.0.1.1	224.0.0.5	OSPF	110	LS Update
22	14.999699929	10.0.1.2	224.0.0.5	OSPF	142	LS Update
23	15.039069918	10.0.1.2	224.0.0.5	OSPF	118	LS Acknowledge
24	15.210042529	10.0.1.2	224.0.0.5	OSPF	158	LS Update
25	15.210820568	10.0.1.2	224.0.0.5	OSPF	122	LS Update
26	15.211424469	10.0.1.2	224.0.0.5	OSPF	94	LS Update
27	15.380527873	10.0.1.1	224.0.0.5	OSPF	90	LS Update
28	15.431807986	10.0.1.1	224.0.0.5	OSPF	90	LS Update
29	15.756586862	10.0.1.1	224.0.0.5	OSPF	398	LS Acknowledge
30	15.915643889	10.0.1.2	224.0.0.5	OSPF	174	LS Update
31	15.916637136	10.0.1.2	224.0.0.5	OSPF	90	LS Update
32	16.766555593	10.0.1.1	224.0.0.5	OSPF	158	LS Acknowledge
33	20.001595451	10.0.1.1	10.0.1.2	OSPF	110	LS Update
34	20.001990730	10.0.1.2	10.0.1.1	OSPF	110	LS Update
35	20.002925331	10.0.1.2	224.0.0.5	OSPF	82	Hello Packet
36	20.003857366	10.0.1.1	224.0.0.5	OSPF	154	LS Update
37	20.004045628	10.0.1.2	224.0.0.5	OSPF	154	LS Update
38	20.004532777	10.0.1.1	224.0.0.5	OSPF	110	LS Update
39	20.006518271	10.0.1.2	224.0.0.5	OSPF	78	LS Acknowledge
40	20.217849552	10.0.1.2	224.0.0.5	OSPF	110	LS Update
41	20.271555537	10.0.1.2	224.0.0.5	OSPF	118	LS Update
42	20.381818257	10.0.1.1	224.0.0.5	OSPF	186	LS Update
43	20.721910888	10.0.1.1	224.0.0.5	OSPF	82	Hello Packet

Frame 1: 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface 0

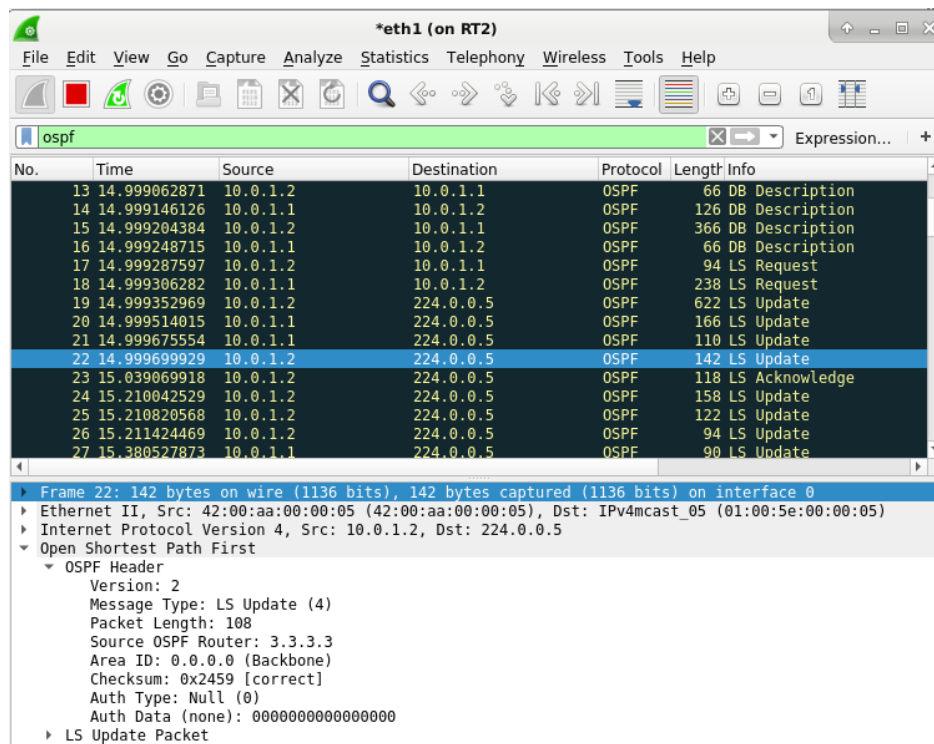
```

0000  01 00 5e 00 00 05 42 00  aa 00 00 05 08 00 45 c0  ..^...B.....E:
0010  00 44 ed 00 00 00 01 59  e0 19 0a 00 01 02 e0 00  .D....Y.....
0020  00 05 02 01 00 30 03 03  03 03 00 00 00 f2 90      ....0.....
0030  00 00 00 00 00 00 00 00  00 00 ff ff 00 00 0a      .....
0040  02 01 00 00 28 00 00 00  00 00 00 00 00 02 02      ....(.....

```

wireshark_eth1_20240205132855_hHVfxw.pcapng Packets: 48 · Displayed: 42 (87.5%) Profile: Default

Na početku vidimo Hello pakete, a kasnije i LS Request i LS update pakete.



No.	Time	Source	Destination	Protocol	Length	Info
13	14.999062871	10.0.1.2	10.0.1.1	OSPF	66	DB Description
14	14.999146126	10.0.1.1	10.0.1.2	OSPF	126	DB Description
15	14.999204384	10.0.1.2	10.0.1.1	OSPF	366	DB Description
16	14.999248715	10.0.1.1	10.0.1.2	OSPF	66	DB Description
17	14.999287597	10.0.1.2	10.0.1.1	OSPF	94	LS Request
18	14.999306282	10.0.1.1	10.0.1.2	OSPF	238	LS Request
19	14.999352969	10.0.1.2	224.0.0.5	OSPF	622	LS Update
20	14.999514015	10.0.1.1	224.0.0.5	OSPF	166	LS Update
21	14.999675554	10.0.1.1	224.0.0.5	OSPF	110	LS Update
22	14.999699929	10.0.1.2	224.0.0.5	OSPF	142	LS Update
23	15.039069918	10.0.1.2	224.0.0.5	OSPF	118	LS Acknowledge
24	15.210042529	10.0.1.2	224.0.0.5	OSPF	158	LS Update
25	15.210820568	10.0.1.2	224.0.0.5	OSPF	122	LS Update
26	15.211424469	10.0.1.2	224.0.0.5	OSPF	94	LS Update
27	15.380527873	10.0.1.1	224.0.0.5	OSPF	90	LS Update

Frame 22: 142 bytes on wire (1136 bits), 142 bytes captured (1136 bits) on interface 0

- Ethernet II, Src: 42:00:aa:00:00:05 (42:00:aa:00:00:05), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
- Internet Protocol Version 4, Src: 10.0.1.2, Dst: 224.0.0.5
- Open Shortest Path First
 - OSPF Header
 - Version: 2
 - Message Type: LS Update (4)
 - Packet Length: 108
 - Source OSPF Router: 3.3.3.3
 - Area ID: 0.0.0.0 (Backbone)
 - Checksum: 0x2459 [correct]
 - Auth Type: Null (0)
 - Auth Data (none): 0000000000000000
 - LS Update Packet